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disproportionates to give oxygen and water. The reaction is, however, slow but is explosive when catalyzed by metal. If it is stored in glass container, it dissolves the alkali metals from the glass, which catalyzes the disproportionation reaction. For this reason, H_2O_2 solutions are stored in plastic bottles.

$$H_2O_2 \rightarrow H_2O + \tfrac{1}{2}O_2$$

Hydrogen peroxide can act both as an oxidizing agent and a reducing agent. Oxidation is usually performed in acidic medium while the reduction reactions are performed in basic medium

In acidic conditions:

$$H_2O_2 + 2\,H^+ + 2\,e^- \rightarrow 2\,H_2O \quad (E^0 = +\,1.77\ V)$$

For example

$$2FeSO_4 + H_2SO_4 + H_2O_2 \rightarrow Fe_2(SO_4)_3 + 2H_2O$$

In basic conditions:

$$HO_2^- + OH^- \rightarrow O_2 + H_2O + 2\,e^-$$

$$(E^0 = +\,0.08\ V)$$

For Example,

$$2KMnO_4\,(aq) + 3\,H_2O_2\,(aq) \downarrow$$

$$2MnO_2 + 2\,KOH + 2\,H_2O + 3O_2\,(g)$$

4.7.3 Uses of hydrogen peroxide:

The oxidizing ability of hydrogen peroxide and the harmless nature of its products, i.e., water and oxygen, lead to its many applications. It is used in water treatment to oxidize pollutants, as a mild antiseptic, and as bleach in textile, paper and hair-care industry.

Hydrogen peroxide is used to restore the white colour of the old paintings which was lost due to the reaction of hydrogen sulphide in air with the white pigment $Pb_3(OH)_2(CO_3)_2$ to form black colored lead sulphide. Hydrogen peroxide oxidises black coloured lead sulphide to white coloured lead sulphate, there by restoring the colour.

$$PbS + 4H_2O_2 \rightarrow PbSO_4 + 4\,H_2O$$

4.7.4 Structure of hydrogen peroxide:

Both in gas-phase and liquid-phase, the molecule adopts a skew conformation due to repulsive interaction of the OH bonds with lone-pairs of electrons on each oxygen atom. Indeed, it is the smallest molecule known to show hindered rotation about a single bond.

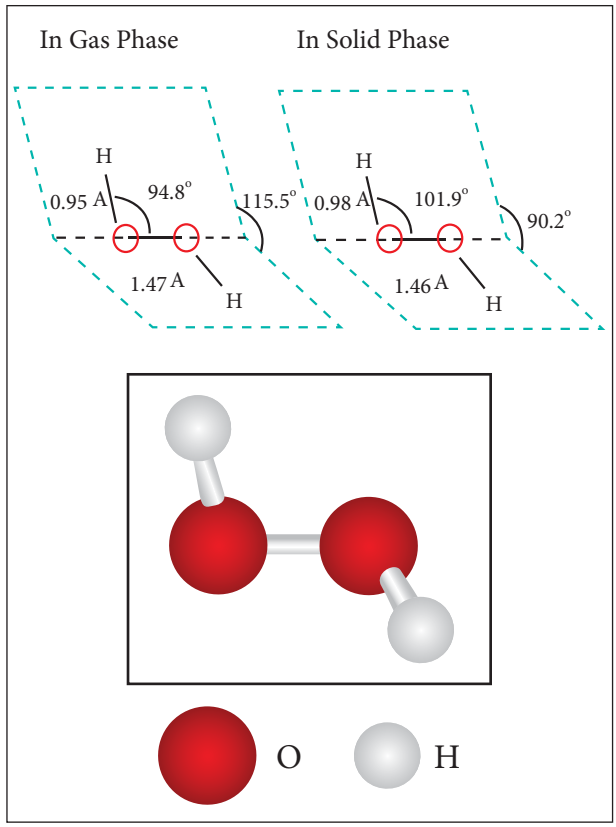

Figure 4.5 structure of H_2O_2